Evaluation Comments on the RSM Proposal

Source reviewed: RSM_paper_main.pdf | Prepared: 4 June 2026

Bottom line

Overall assessment: the core idea is promising and the common-loading theory is a real contribution. However, the current draft should not yet be treated as submission-ready. The main risks are overclaiming around the power-law extension, the interpretation of alpha, the endogenous Monte Carlo link, and the SPF application.

1. Executive verdict

The proposal has a clear and useful object: bagged random-subset forecast combination. The exact finite-bagging bridge, the common-loading derivation, and the weight-deviation representation create a coherent theoretical core. The paper's main story - that bagging changes the excess-risk decay and therefore changes the optimal subset-size scaling - is attractive and worth developing.

The current draft is best viewed as a strong working paper/proposal rather than a finished submission. Some headline claims are stronger than what is currently proven, and several empirical interpretations need to be qualified.

Most urgent correction: the paper currently risks saying that near-homogeneous forecasters correspond to alpha about 1. The theory implies the opposite conditional on a non-negligible excess-risk curve: small heterogeneity gives alpha about 2. Near-homogeneity mainly means that C_delta is close to zero, so the curve is flat and alpha is weakly identified.

2. What works well

The finite-bagging bridge is clean and useful.

The identity separating infinite-bagged population risk from the finite-G residual subset variance is exact and valuable:

```
Q_k^{bag,G} = Q_k^{bag,inf} + (1/G) * (Q_k^{sub} - Q_k^{bag,inf})
```

This gives a transparent interpretation: finite bagging leaves a 1/G fraction of the average-subset gap. It also makes clear that bagging is not only a computational device; it changes the theoretical object.

The common-loading model is a strong theoretical laboratory.

With Sigma_e = q ii' + D, the common component cannot be diversified away under sum-to-one weights. The subset risk simplifies to:

```
Q(S) = q + 1/A_S
```

That simplification is elegant and gives the paper a tractable environment in which one can derive exact and approximate expressions for the RSM risk.

The weight-deviation representation is the strongest theoretical result.

```
R_k^{bag,inf} = Q_k^{bag,inf} - Q^{opt} = sum_i (vbar_{i,k} - w_i^{opt})^2 / tau_i
```

This formula is conceptually powerful. It shows that the excess risk of bagging is exactly a weighted squared distance between the average bagged RSM weight vector and the oracle weight vector. The vanishing of the linear term is not an approximation; it follows from the first-order condition and the adding-up constraint.

The small-heterogeneity expansion is convincing.

```
vbar_{i,k} - w_i^{opt} ~= - delta_i/(m-1) * (1/k - 1/m)
```

```
R_k^{bag,inf} ~= C_delta * (1/k - 1/m)^2
```

This is a real derivation, not just an imposed C/k^2 assumption. The endpoint correction also matters because it respects the exact boundary condition at $k = m$.

3. Biggest theoretical problem: the alpha interpretation

The power-law section currently appears to contradict the earlier theory. Under small heterogeneity, the draft derives:

```
R_k^{bag,inf} ~= C_delta * z_k^2, where z_k = 1/k - 1/m
```

That corresponds to alpha = 2 in a regression of log R_k on log z_k, provided the excess-risk curve is large enough to identify the slope. In contrast, the dominant-forecaster or extreme-heterogeneity limit can generate an effective slope closer to alpha = 1, depending on the fitting range.

Therefore the natural theoretical ordering is:

```
small heterogeneity     -> alpha about 2, if identifiable
extreme dominance       -> alpha closer to 1
near homogeneity        -> C_delta about 0, slope weakly identified
```

The draft's Monte Carlo table reports alpha about 1.1 at H = 1.05 and alpha about 1.7 at high heterogeneity. I would not interpret the H = 1.05 slope literally. In that case the excess-risk curve is almost flat, so log-log slope estimates can be unstable, especially if R_k is very close to numerical or Monte Carlo noise.

> Recommended replacement language: Near-homogeneity means C_delta is close to zero, so R_k is nearly flat and alpha is weakly identified or practically irrelevant. Conditional on a non-negligible excess-risk curve, the small-heterogeneity expansion predicts alpha about 2.

4. Inflation term: useful but too strong as currently stated

The interior optimum depends on combining the decreasing population approximation error with the finite-sample inflation factor:

```
I(T,k) = (T - 1)/(T - k)
```

This is a reasonable first-order approximation for a constrained OLS estimator with k - 1 free parameters. But the paper applies it to bagged RSM. That is plausible as a conservative device, yet it is not an exact degrees-of-freedom calculation for the bagged estimator.

A reviewer is likely to ask:

- Why should bagged RSM have the same inflation as one size-k constrained OLS estimator?
- Does the effective degrees of freedom depend on G, m, subset overlap, and the covariance structure?
- If the inflation is conservative, does it bias predicted k* downward or upward?
- Why does the endogenous Monte Carlo show k*(L_bag) = 26-32 while the RMSFE-optimal k is only 11-14?

The last point is the most damaging. The draft's own endogenous results show that the inflation-based criterion overpredicts the RMSFE-optimal subset size. Therefore, the cube-root rule should be presented as a theoretical guide under a maintained inflation approximation, not as a precise empirical rule.

5. Endogenous Monte Carlo: define the estimator more carefully

The theory defines subset weights using the inverse covariance matrix of the selected subset:

```
w_S = Sigma_{e,S}^{-1} i_S / (i_S' Sigma_{e,S}^{-1} i_S)
```

But the endogenous Toeplitz and factor simulations appear to use diagonal precision weights of the form tau_i/A_S. That is optimal under the diagonal/common-loading structure, but it is not the optimal constrained minimum-variance weight for general Toeplitz or factor covariance matrices.

You have two defensible options:

- Use full subset inverse-covariance weights in the Toeplitz and factor simulations, matching the theoretical definition.
- Or explicitly rename the simulated estimator as a diagonal-precision approximation to RSM, not the exact estimator from Definition 1.

Right now the draft moves between those interpretations. That ambiguity will confuse readers and give reviewers an easy target.

6. SPF application: currently weak as evidence for the theory

The SPF section says that at each round the procedure draws k forecasters and averages their forecasts. But if subset forecasts are equally averaged, then infinite bagging mechanically gives the full equal-weighted mean:

```
E_S[ (1/k) * sum_{i in S} f_i ] = (1/N) * sum_{i=1}^N f_i
```

Therefore, if no estimated precision or regression weights are used, the SPF version of infinite-bagged RSM is not a substantive alternative to the mean. Any tiny differences with G = 1000 are likely finite simulation noise, changing panel composition, or implementation details.

This does not mean the SPF section is useless. It can be a good sanity check showing that when the method collapses toward equal averaging and forecaster heterogeneity is limited, it is hard to beat the mean. But it should not be presented as strong validation of the main weighted-RSM theory.

> Also check the sign convention in the SPF gain column. If gain is defined as (L_EW - L_RSM)/L_EW, then cases where RSM has a lower RMSFE than the mean should have positive gains, not negative ones.

7. FRED-MD empirical section: distinguish fixed-k from oracle-k results

The FRED-MD application reports both default k = floor(sqrt(T)) results and ex post optimal k* values such as 19 or 21. The default-k results are cleaner as out-of-sample evidence. The ex post k* values are useful diagnostics, but they should not be treated as genuine out-of-sample performance unless k is chosen using only past data.

A stronger empirical structure would be:

- Main performance: fixed rule, such as floor(sqrt(T)), floor(T^(1/3)), or a pre-specified theory-guided rule.
- Oracle diagnostic: ex post RMSFE(k) curve to show whether the loss surface is U-shaped and where the evaluation-sample optimum lies.
- Optional real-time rule: choose k recursively from a rolling or expanding validation window.

This would protect the empirical section from a data-snooping criticism while still allowing you to show the economically interesting U-shaped curve.

8. Claim-level edits

The current abstract and conclusion should be made more precise. Suggested changes:

- Replace 'exact functional form' with 'exact representation' plus 'closed-form small-heterogeneity approximation'.
- Replace 'strictly more efficient than the square-root rule' with 'implies a lower subset-size scaling rate under the maintained inflation approximation'.
- Replace 'consistent with the theory' for the endogenous case with 'consistent with the extended estimation-noise interpretation'.
- Remove or correct the phrase 'Propositions 9-??'.

- Avoid implying that RSM uniformly dominates all benchmarks in FRED-MD. In some tables, Lasso or Ridge is competitive depending on the metric and sample.

9. Suggested revised main message

> We show that, under a common-loading forecast-error structure, infinite bagging transforms random-subset forecast combination into a deterministic average-weight estimator whose excess risk is exactly a weighted squared distance from oracle weights. Under small precision heterogeneity, this distance is quadratic in 1/k - 1/m, implying a cube-root subset-size rule under a standard finite-sample inflation approximation. Simulations confirm the exact identities and show that the qualitative interior-k trade-off survives under estimated covariance and non-common-loading structures. Empirically, RSM is useful in heterogeneous high-dimensional macro panels but offers little advantage in near-homogeneous SPF panels.

10. Priority revision checklist

Priority	Revision	Why it matters
1	Fix the alpha interpretation.	Small heterogeneity implies alpha about 2 when identifiable; near-homogeneity mainly means C_delta is near zero, not alpha about 1.
2	Recompute or qualify alpha estimates in the flat low-H regime.	When excess risk is almost zero, log-log slopes are weakly identified and can be driven by numerical or simulation noise.
3	Clarify the endogenous estimator.	Toeplitz and factor Sigma_e require full subset inverse-covariance weights if you want to match Definition 1.
4	Downgrade the inflation factor to a maintained approximation.	The cube-root rule depends on this assumption; bagging may have lower effective degrees of freedom than k.
5	Reframe the SPF application.	If subset forecasts are equally averaged, infinite bagging mechanically recovers the full equal-weighted mean.
6	Separate fixed-k performance from oracle k* diagnostics.	Ex post k* selected on the evaluation sample is descriptive unless chosen using only past data.
7	Tighten claims in the abstract and conclusion.	Use exact, approximate, heuristic, and empirical claims in their proper hierarchy.

11. Final assessment

The proposal has a real contribution. The common-loading derivation is valuable, and the bagged-risk representation is a strong foundation. But the draft currently mixes exact theory, small-heterogeneity approximation, empirical power-law fitting, and endogenous covariance estimation too freely.

The paper will become much more convincing if it imposes the following hierarchy:

```
Exact under common loading
  -> small-heterogeneity approximation
  -> heuristic/general power-law extension
  -> simulation and empirical evidence
```

Once the alpha interpretation, SPF section, endogenous-weight definition, and fixed-k versus oracle-k empirical distinction are corrected, the proposal will look much more coherent and defensible.